

INTRODUCTION TO ECONOMICS

MGT404

Course Introduction and Market Demand

Yale

Introduction (1): Economics

Philadelphia

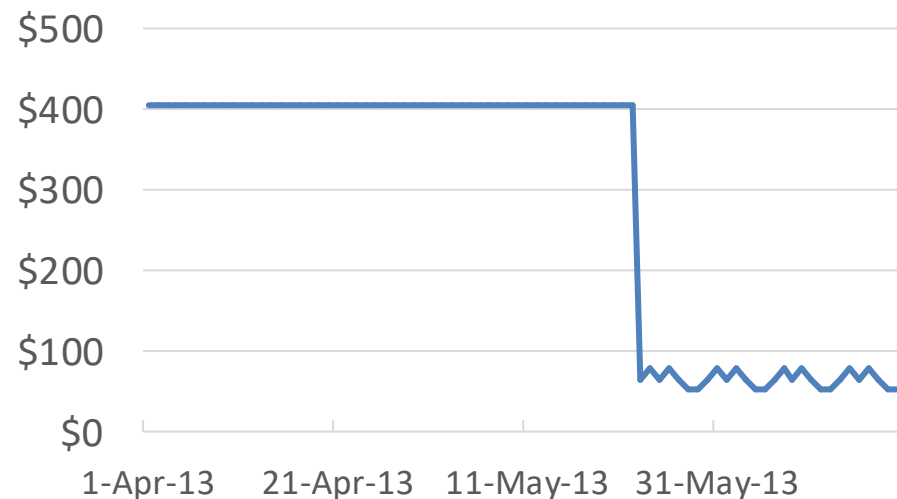


Boston



Jan 2013: Only airline offering nonstop service

One-Way Airfare PHL-BOS¹



- Airfare falls from constant \$404 to a range of \$54-\$79 on May 23.
- JetBlue started 5 daily non-stop flights on the route on May 23.
- Capacity increases, price must fall to fill seats, but... what other incentives exist?

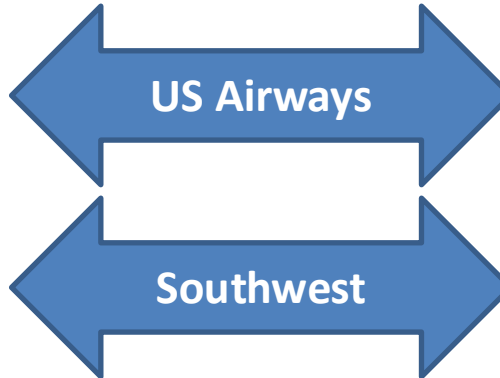
¹ Fares provided by ITA Software

Introduction (1): Economics

Philadelphia



Boston



4Q 2009

- US Airways is the only carrier flying direct BOS- PHL
- Average fare: \$346.90¹

3Q 2010

- Southwest launches competing direct service
- Average 4Q fare: \$121.92

1Q 2012

- Southwest drops direct service
- Average 1Q fare: \$321.22

Was the 2010-2011 price drop due to just increased competition?
Or, was US Airways using a “War of Attrition” to get Southwest to exit the route?

¹ All fares from DOT Consumer Air Fare Report

Introduction (2): Economics

(Another) war in Afghanistan:

How the US military's opium war in Afghanistan was lost

By Justin Rowlett
BBC News

25 April 2019



The US has spent \$1.5m (£1.15m) a day since 2001 fighting the opium war in Afghanistan. So why is business still booming?



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DoD Report: Afghanistan Could Become A 'Full-Fledged Narco State'



Richard Sisk, Military.com Nov 21, 2013, 10:49 AM



Recent reports suggest that the Taliban has won the war on drugs in Afghanistan, the source of 75 percent of the world's heroin.

Despite the \$7 billion U.S. anti-drug effort, Afghanistan has produced a record poppy crop this year and was in danger of becoming a "full-fledged narco state" with most of the profits going to fund the Taliban, the U.N. Office of Drugs and Crime said in a report last week.



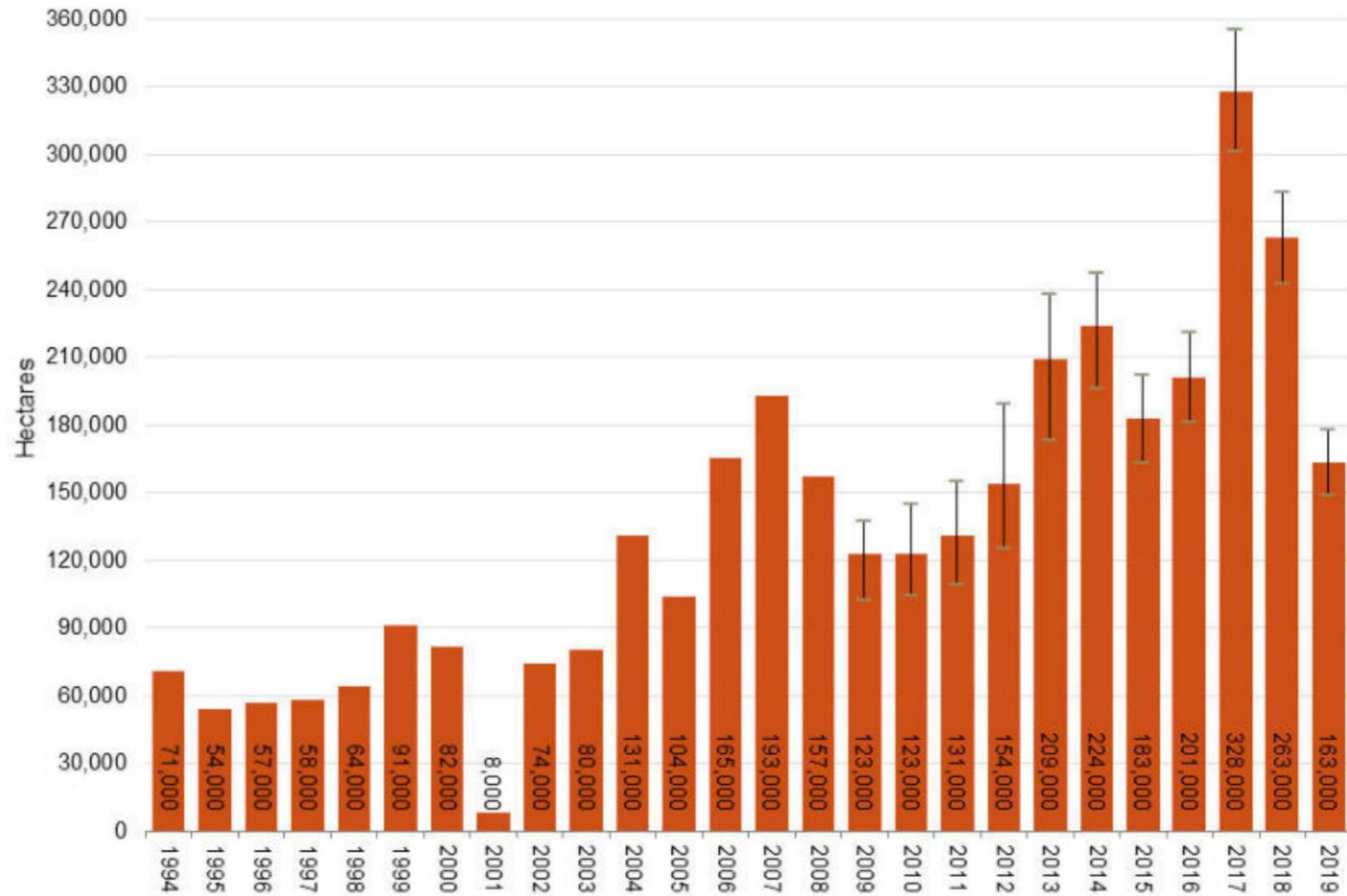
A U.S. soldier of the 101st Airborne Division destroys opium poppies growing in a field during a patrol with Afghan police in Tani district of Khost province, Afghanistan. AP



Since mid-2000's, the US has been fighting poppy cultivation in Afghanistan.

- One of Taliban's main funding sources.
- Accounts for 75% of the world's heroin.

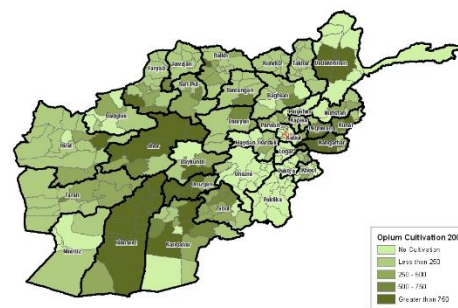
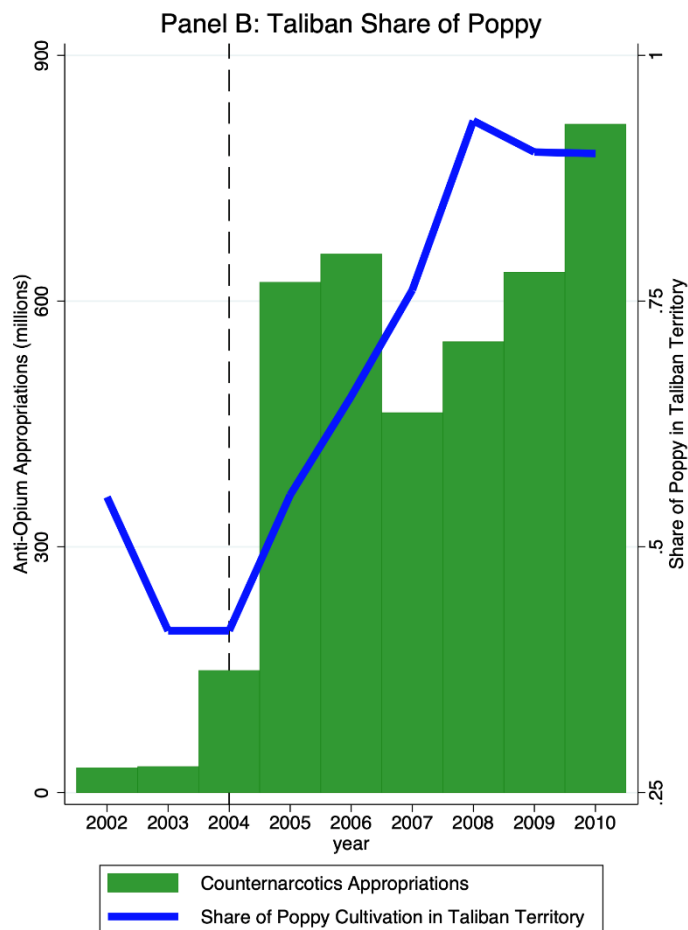
Introduction (2): Economics



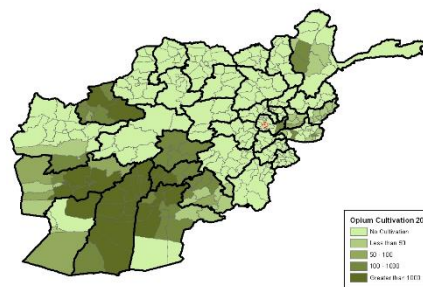
Sources: UNODC/NSIA/MCN opium surveys 1994-2019. The vertical lines represent the upper and lower bounds of the 95% confidence intervals.

Introduction (2): Economics

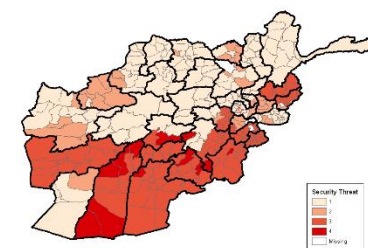
To make things even worse...



Opium Cultivation in 2004



Opium Cultivation in 2009



Security Threat

Jeffery Clemens, 2013, Evaluating Economic Welfare: Lessons from Efforts to Suppress the Afghan Opium Trade, Working Paper.

Introduction (2): Economics

So what's wrong?

- Costs of poppy seeds only account for a small portion of the total cost of retailers:
 - Changes in price do not significantly affect poppy seed demand.
- Can only enforce policies in government-controlled area.
 - Reduce competition for poppy seed farmers in Taliban-controlled areas.
 - Low entry barrier: “More of *Bake Off* than *Breaking Bad*”.

Introduction (3): Economics



- Zillow stock dropped 45% between 11/01/21 and 11/05/21.
- Nasdaq: $\approx 0.02\%$ change in the same period
- What happened?



Introduction (3): Economics

How it started

Zillow Intends to Buy and Flip Homes

Online listing service's strategy threatens to disrupt traditional real-estate-agent business model

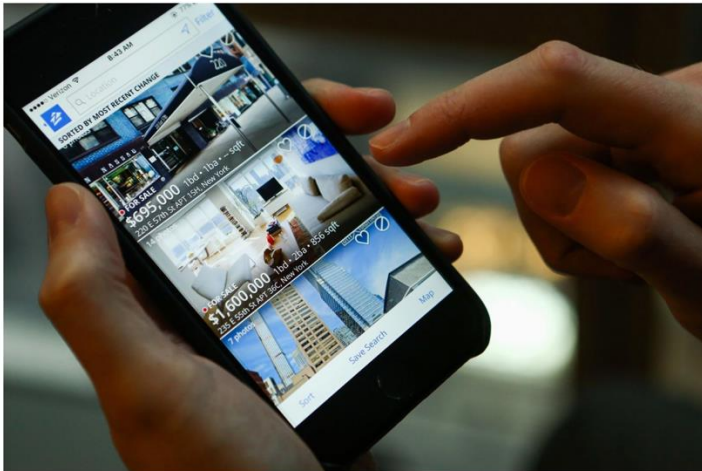
By [Laura Kusisto](#) [Follow](#) and [Rolf Winkler](#) [Follow](#)

April 12, 2018 9:01 pm ET

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Zillow plans to buy, renovate and flip homes in Las Vegas and Phoenix within 90 days. PHOTO: CHRIS GOODNEY/BLOOMBERG NEWS

How it's going

Zillow, facing big losses, quits flipping houses and will lay off a quarter of its staff.

The real estate website had been relying on its algorithm that estimates home values to buy and resell homes. That part of its business lost about \$420 million in three months.

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Zillow's homebuying program

- Started in 2018.
- The plan: Be the “market maker”—buy houses, quickly flip them, and then sell.
- The hope: Go big, and profit from essentially the “transaction fees.”
- Confident in pricing thanks to Zestimate.

Introduction (3): Economics

What went wrong?

- Holding house inventory is costly...Unlike holding stocks.
 - Property taxes
 - Utilities (for inspection, temperature/humidity control)
 - Other fees (e.g., HOA fees)
 - Timeframe much longer compared to selling stocks—More uncertainty!
 - Who could have predicted the post-COVID boom?
- Asymmetric information!
 - House sellers and local buying competitors know much more than Zillow's algorithm.
 - Skeletons in the closet? Which side of the street is better? Do neighbors judge you for not mowing your lawn?
 - **Conditional on** Zillow winning the bid and gets the house, what kind of house do you think it's more likely to be?

This course has two main take-aways:

Incentives

- Agents (consumers, managers, firms, etc) are driven by their incentives.
- Understanding an agent's incentives means understanding what they will do in any situation.
- We will develop rigorous tools to model/analyze incentives

Equilibrium

- Given a system of Agents, we can surmise what will happen based on the notion of an equilibrium.
- This is a state where no agent would want to change their behavior.

Once you understand incentives and outcomes, you can make better decisions.

Why MGT404?

Goal of this class:

This class is designed to equip you with the toolset required to analyze complex business situations



Applications begin in other fields of Business:

Marketing

Pricing, Market Power, Oligopoly

Operations

Cost Minimization

Finance, Real Estate

Equilibrium Analysis, Uncertainty

Management

Externalities, Asymmetric Information

Our Approach:

This class takes a **rigorous** approach – we will set up models of consumers and firms, and solve for what rational agents will do. The models involve algebra and calculus, but results in a deeper understanding of the forces that shape the business world.

Microeconomics at Yale SOM

Yale SOM uses a “team teaching” approach.

MGT 404: Microeconomics



Professor
Kai Hao Yang



Professor
Barbara Biasi

Game Theory

Professor
Jidong Zhou



MGT 410: Competitor



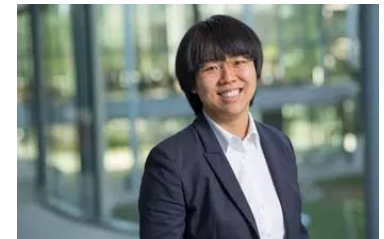
Professor
Kevin Williams



Professor
Kai Hao Yang



Professor
Barry Nalebuff



Professor
Zeqiong Huang

Fall 2025

Electives beyond: Competitive Strategy, Behavioral Economics, others.

My Personal Background and Research

Professor Kai Hao Yang

kaihao.yang@yale.edu

Research Interests: Economic Theory

- Mechanism/Market Design
- Game Theory & Information Economics
- Digital Economics
- Market Structure
- Political Economy

- Ph.D. in Economics from the University of Chicago.
- Postdoc at the Cowles Foundation.
- 5th year at Yale SOM.
- Recent research topics: privacy, algorithmic fairness, discrimination, gerrymandering, security design, data broker, price discrimination, influencer economics, online platforms
- Ask me about research if you're interested!

Course Overview

Sessions

1-4

**The context in which firms
operate**

- Demand and Supply
- Equilibrium and Elasticity
- Government interventions

5

Production decisions

- Cost minimization
- Perfectly competitive markets
- Producing with market power

6-9

Pricing

- Individual and self-selecting price discrimination
- Quantity discounting
- Complex pricing schemes

10-13

Information Econ

- Risk and uncertainty
- Moral hazard
- Adverse selection

14

Review session

A Note on Math Requirements

We expect that you are “familiar” with basic algebra and calculus:

- Manipulation of functions, solving equations, taking derivatives (c.f., math lecture on Canvas).
- Not trying to torture you with math, but mathematical models are important tools in economics.
- TAs are here to help.

My advice on staying on top of things:

- **READ** problem sets as we cover relevant material.
- **ATTEMPT** problems on your own, before talking to your team or others.
- **COME PREPARED** to learning team meetings regarding problem sets.
- **REVIEW** when you found that you had gone down the wrong road.
- **NOTE** topics that are challenging you, so that you can do extra practice.



You come from a diverse background. To some, this will be easy. To others, difficult.
However, we only admit students we believe can do this.

Grading

	Due Date	% Of Grade
Problem Set 1	September 12 (noon)	10%
Problem Set 2	September 26 (noon)	10%
Problem Set 3	October 8 (noon)	10%
Exam	October 13 (3hrs)	60%
Participation	-	10%

Problem sets:

- Work together with your learning team to solve the problem sets.
- Each person must submit their own final problem set.
- Problem Sets are due online via Canvas. No time limits once you begin.
- No partial credit for incorrect answers.

Exam:

- **Closed book**, except for a one-page, letter-sized (doubled sided) note.
- Partial credit will be given.

AI for learning: Sure!

AI for problem sets/exams: Not allowed!

(Be aware that it makes many simple mistakes)

Materials

Files

Search files

Search files...

Enter at least 2 characters to search

MGT 404 (FA25): Basics of Economics/Game Theory > Basic of Econ > Class readings

☐	Name ^	Created ↕	Last Modified ↕	Modified By ↕
☐	 Review Session	Oct 10, 2023	Thursday	
☐	 Session 1	Aug 23, 2022	Thursday	
☐	 Session 2	Aug 23, 2022	Thursday	
☐	 Session 3	Aug 23, 2022	Thursday	
☐	 Session 4	Aug 23, 2022	Thursday	
☐	 Session 5	Aug 23, 2022	Thursday	

Course materials are available on Canvas: Readings, problem sets, etc.

- Typically 2-3 readings (articles) per session.
- Three problem sets:
 - Print out PDF, complete on your own, enter solutions via Canvas.
- Readings from textbook are optional (and coordinated with lectures).

Classroom Norms

Participation

- The expectation is that all students have read the articles assigned for each class and will be prepared to discuss.
- Most of all: be present. Contribute your expertise.

Electronics / Remote Students

- Limit distractions: close any applications that give notifications (e.g. Outlook), put phone on silent mode, etc.
- Please give me feedback (can be anonymous via class reps).



We want you to be engaged, we want everyone to learn the way they learn best,
and we will design lessons assuming you are prepared.

Please, please, please tell me if I am mis-pronouncing your name.

Today's Plan

What is demand:

1. Reservation Prices

- The *most* a consumer is willing to pay for a good or service

2. The Demand Curve and Consumer Surplus

- Reflects the *distribution* of reservation prices in a market: at a given price, how many are willing to buy?

3. Aggregating Demand

- When adding demand from different types of consumers, add in *quantities*.

4. Questions and discussion

- Application to TSA PreCheck

Reservation Price

Definition: Reservation Price

A consumer's **Reservation Price** for a good or service is the highest price at which they would be willing to purchase the good or service.



This is not what you would *like* to pay; it is the most you would be *willing* to pay.

A consumer's reservation price can change based on characteristics such as time, income, and quantity.

Basic concept extends to supply, wages, etc.

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Constructing a demand curve

What is your **reservation price** for:

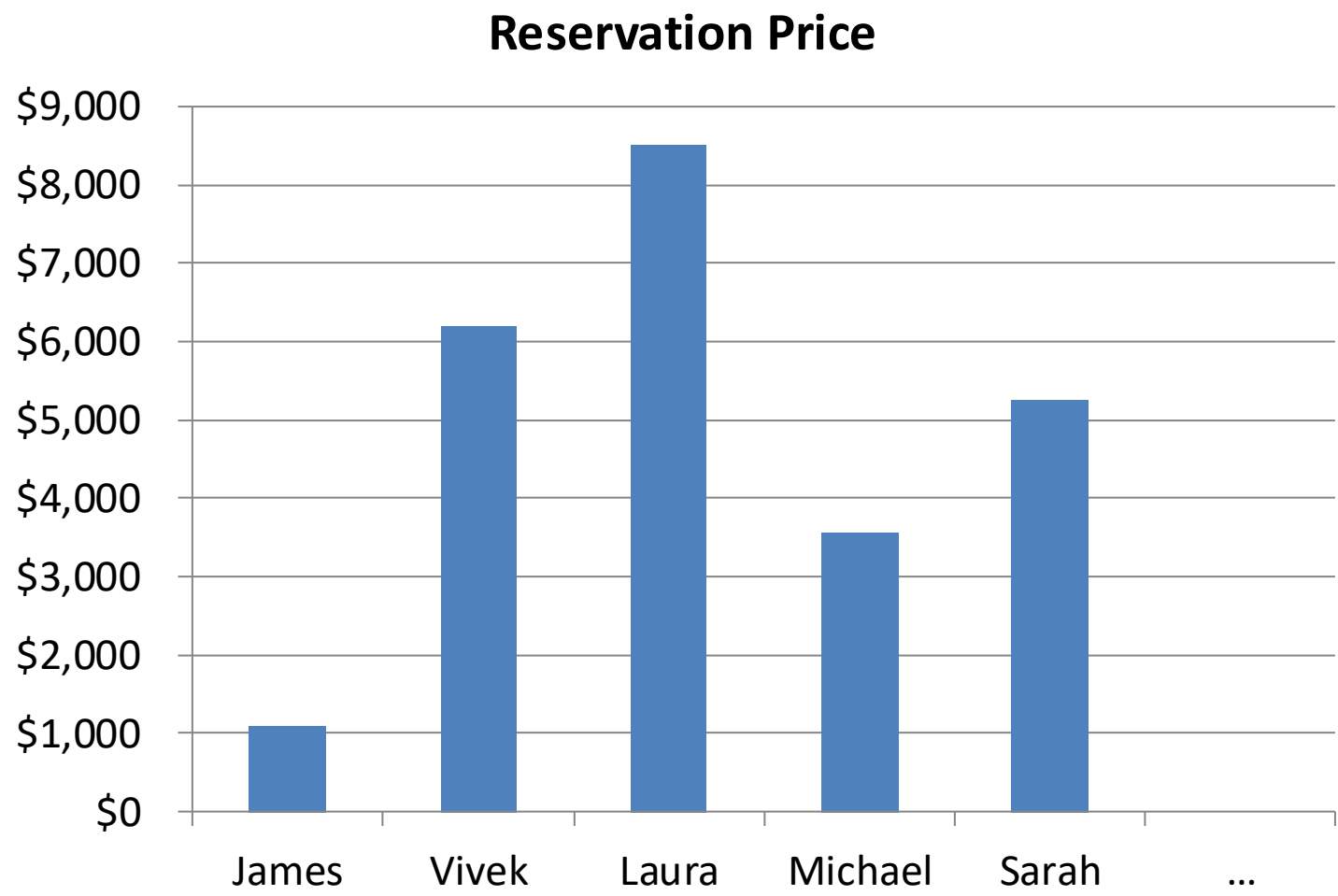
A private jet vacation (any week you want) to an isolated Caribbean isle for one week, for you and your friends/family (you + 5)

<https://forms.gle/5UhPekdQfnDLvAN26>

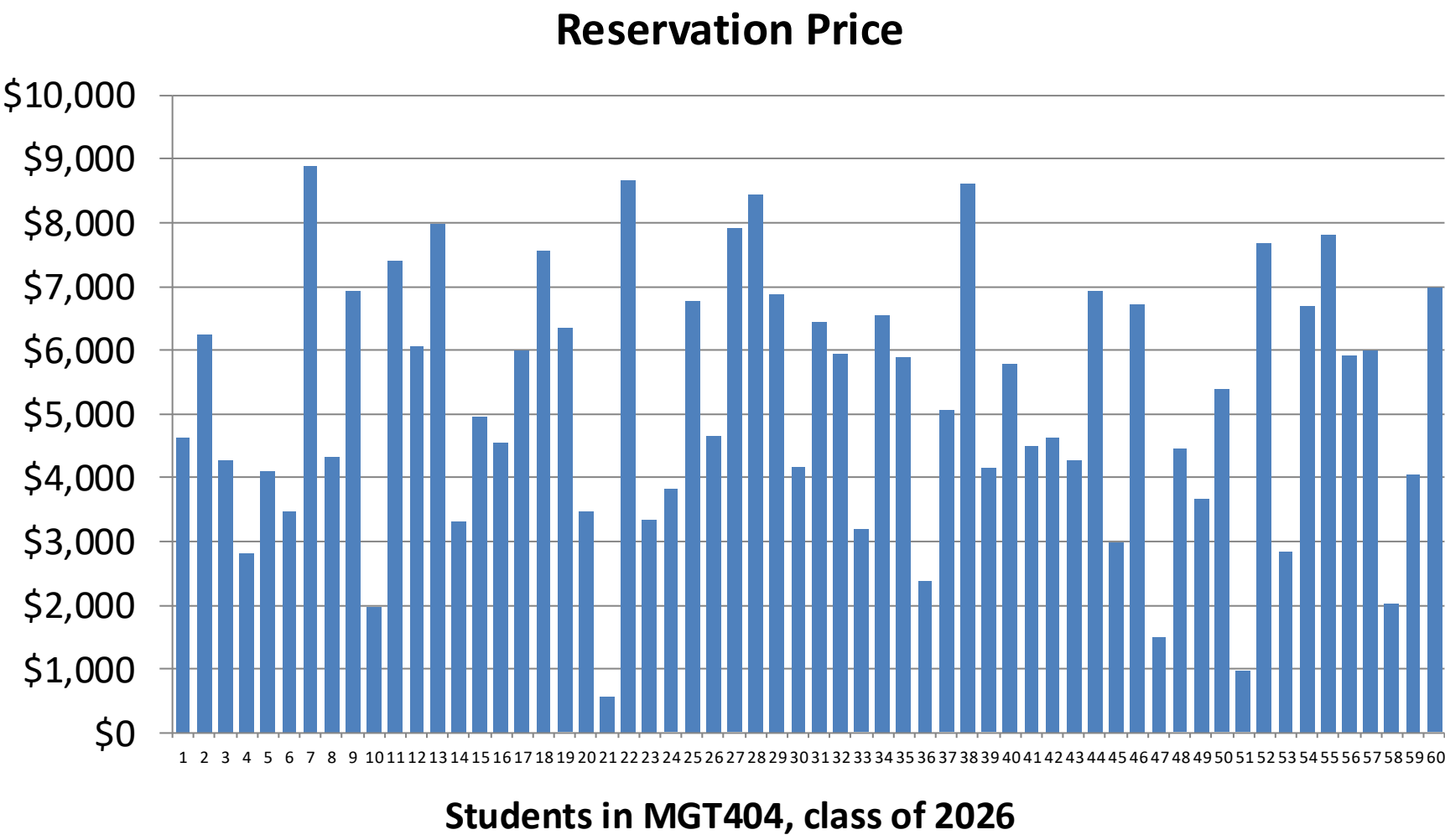
(case sensitive)



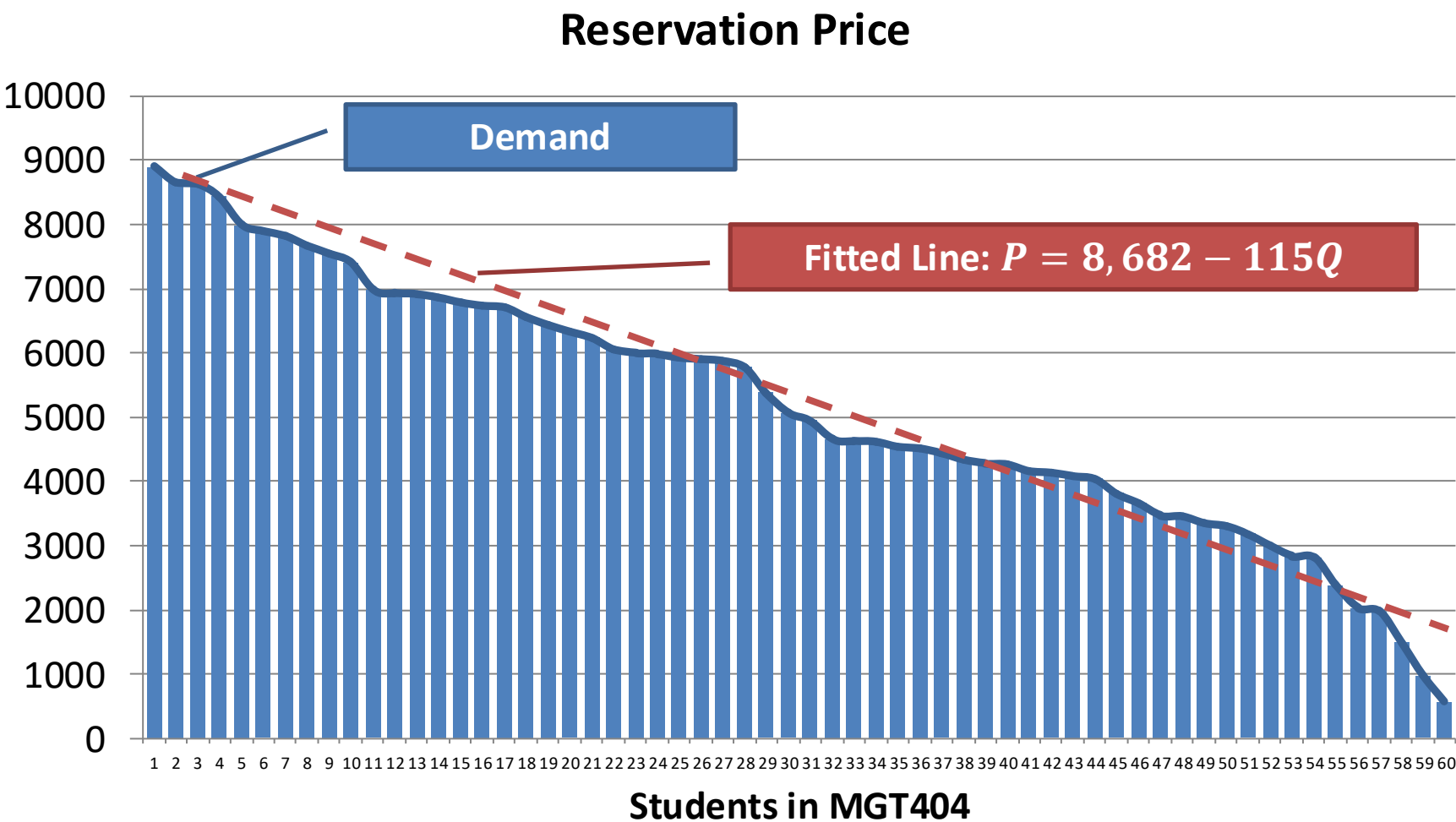
Demand from Reservation Prices



Demand from Reservation Prices



Demand from Reservation Prices



Aside: “Demand” vs “Inverse Demand”

Note that we can write the demand function in two different, but equivalent ways:

$$P = 100 - 2Q$$



$$Q = 50 - 0.5P$$

- Great for graphing/plotting as we put the Price on the Y-axis and Quantity on the X-axis

- Can be more intuitive as “demand” is thought of as a quantity
- Clear to see that if you increase price, quantity must fall.



Sometimes referred to as
“inverse demand”

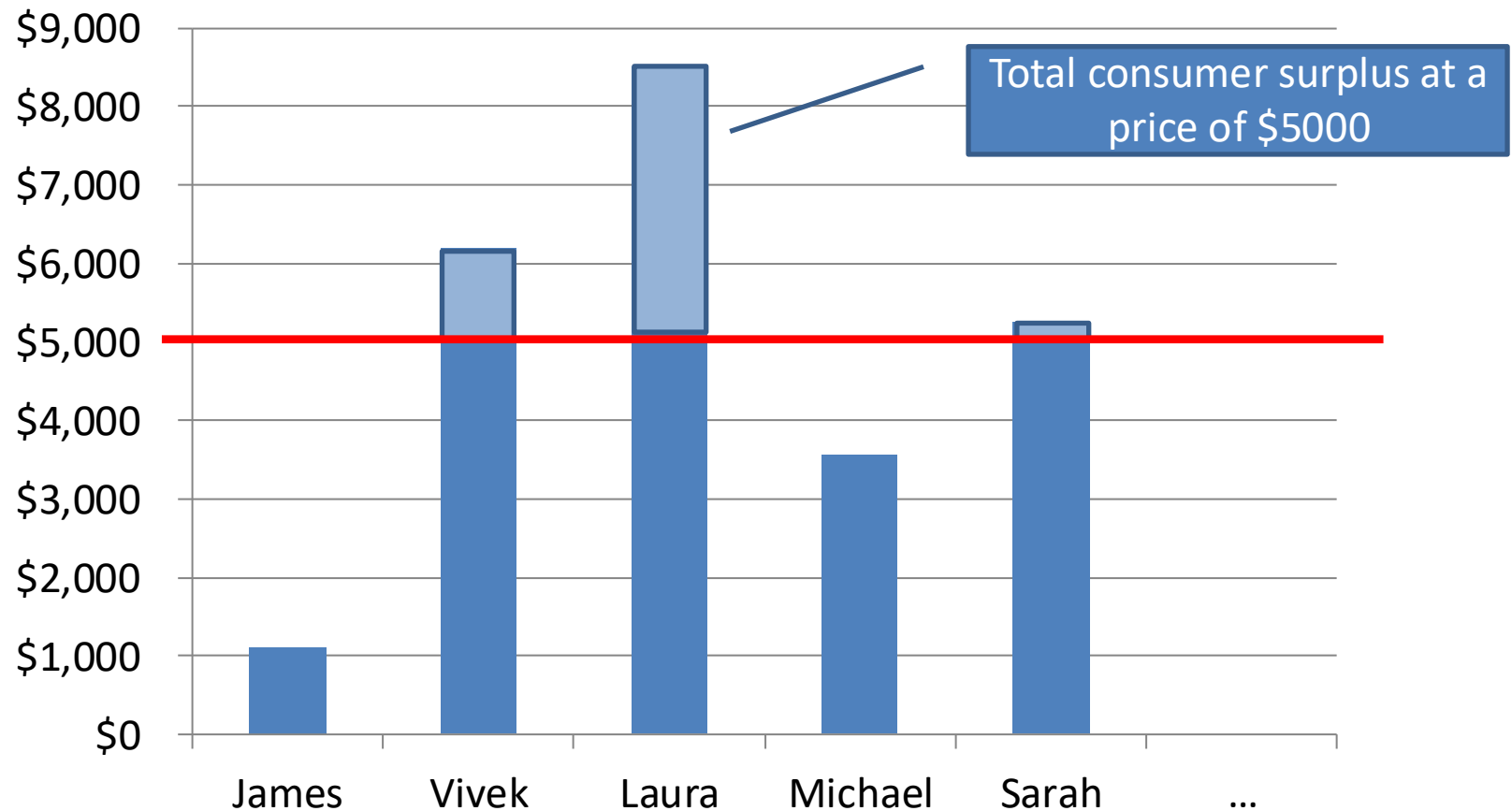


Usually referred to as “demand”

Consumer Surplus from Reservation Prices

Consumer Surplus: the difference between my reservation price and the price I pay. A measure of how “well off” consumers are.

Reservation Price: Private Jet Trip

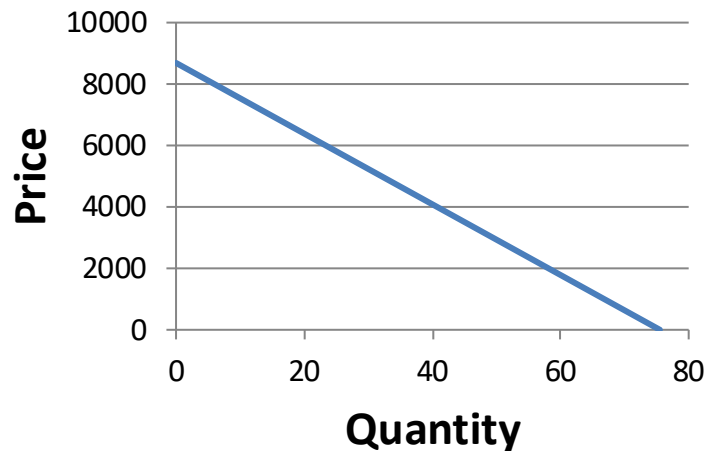


Consumer Surplus in General

Recall: EMBA Demand for the Vacation

We estimated demand for the private jet vacation to be

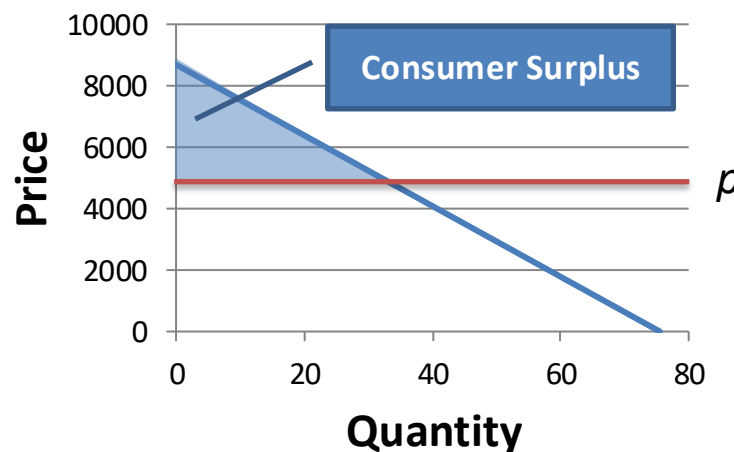
$$P = 8,682 - 115Q$$



Suppose I price the private jet trip at \$5,000. What is consumer surplus?

Consumer Surplus

We can solve for consumer surplus by calculating the area between demand and price:



Use the demand equation to find base and height of triangle.

Consumer Surplus at a price of \$5,000:

$$\frac{1}{2}(\text{base})(\text{height}) = 0.5 * (32) * (3,682) = \$58,912$$

How can we compute a demand function?

Method	Considerations
Survey/Focus Group	• Ask people their “reservation price”. Plot the distribution, and you get a demand curve. • How reliable are answers?
Conjoint Analysis	• Consumers choose between subsets of products/prices. • Use stats methods to get at willingness to pay. • You can learn this in marketing classes.
Regression	• Works well with commodities (undifferentiated goods) • Regress quantity on price for different times/markets. • BUT—these changes must be caused by shifts in supply. If demand is shifting and causing the different prices and quantities, you’re not moving along a single curve
Experiments	• Change the price or features in different markets or to different consumers and measure the reaction. • Can be hard to implement. See: Amazon.com.
Auctions	• The Becker-DeGroot-Marshak auction is used frequently by economists—ask people to name a price, then draw a number. If below given price, transaction happens at the drawn price.

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3. Aggregating Demand

- **When adding demand from different types of consumers, add in *quantities*.**

4. Questions and discussion

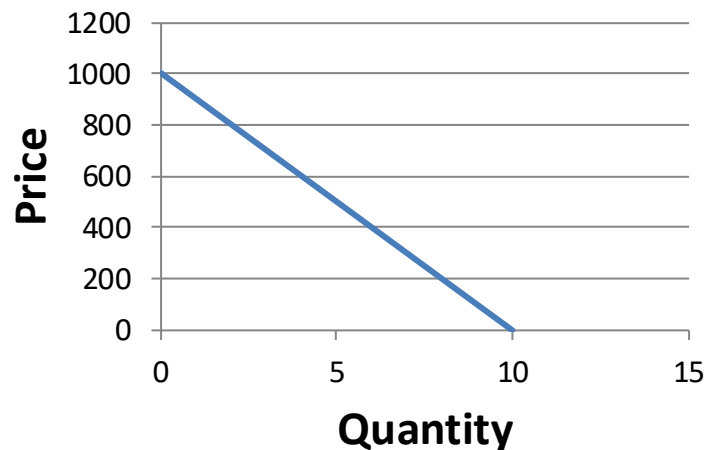
- Application to TSA PreCheck

Aggregating Demand

Demand for Vacation from New Haven MBAs

We estimated demand from New Haven-based MBAs:

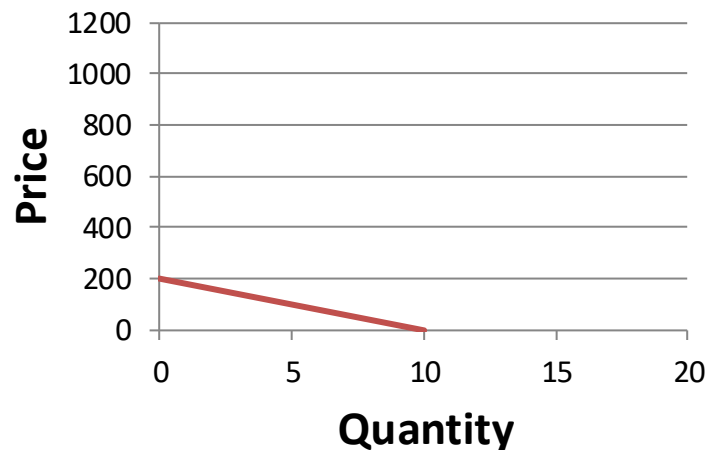
$$P = 1000 - 100Q$$



Demand for Vacation from Miami MBAs

Separately, we estimate demand among those from Miami:

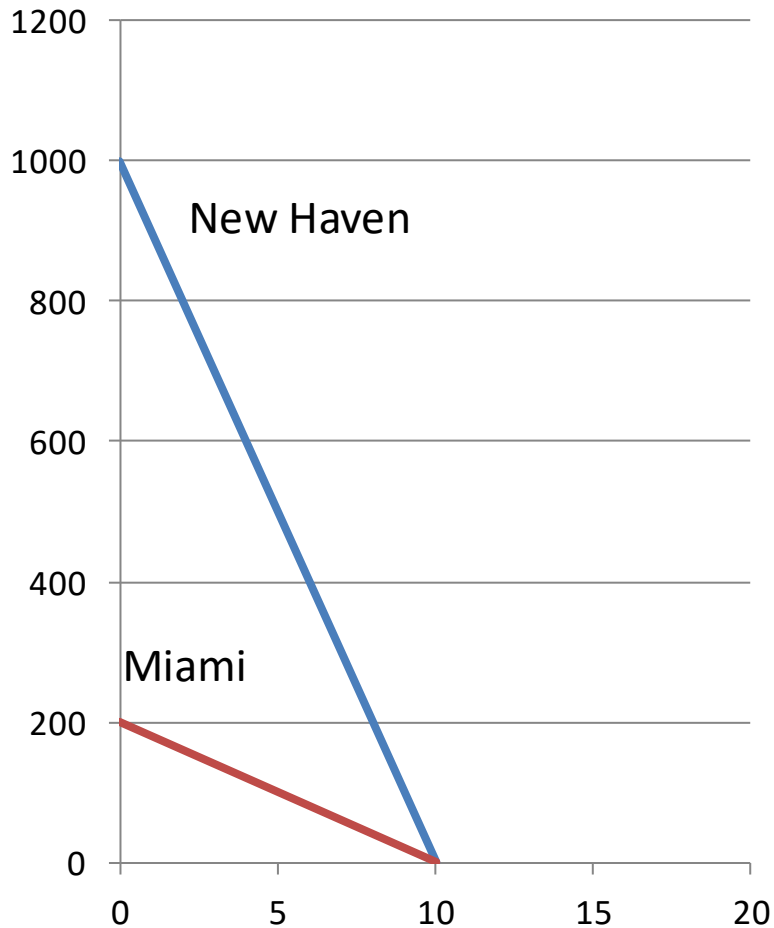
$$P = 200 - 20Q$$



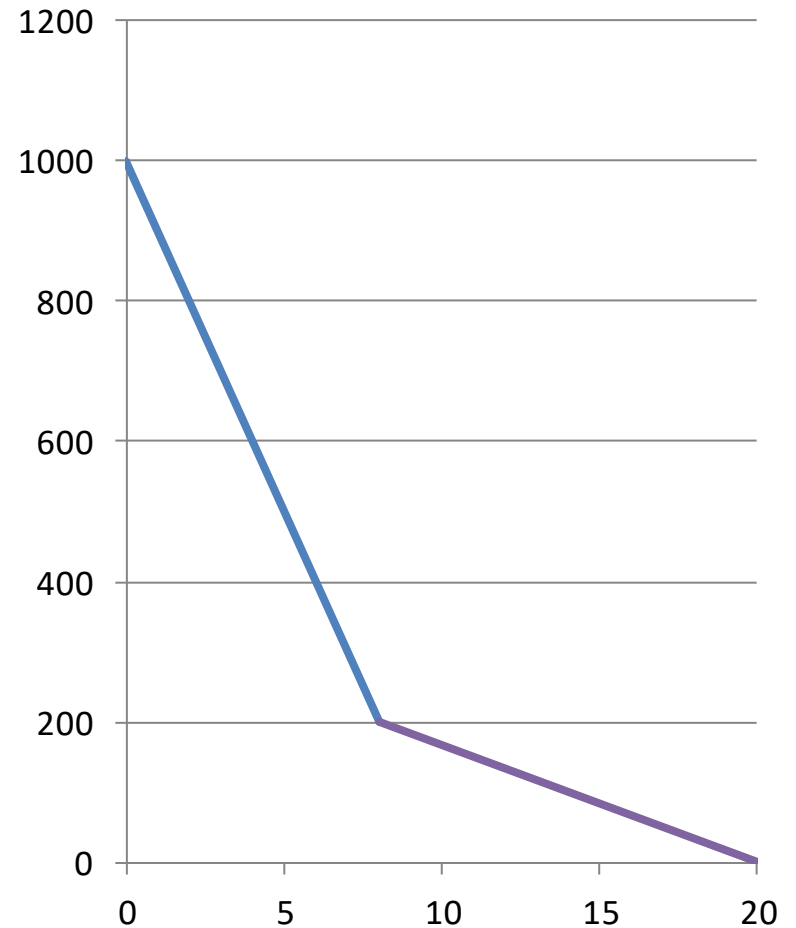
What would be the **combined** demand function for both groups?

Aggregating Demand: Graphic

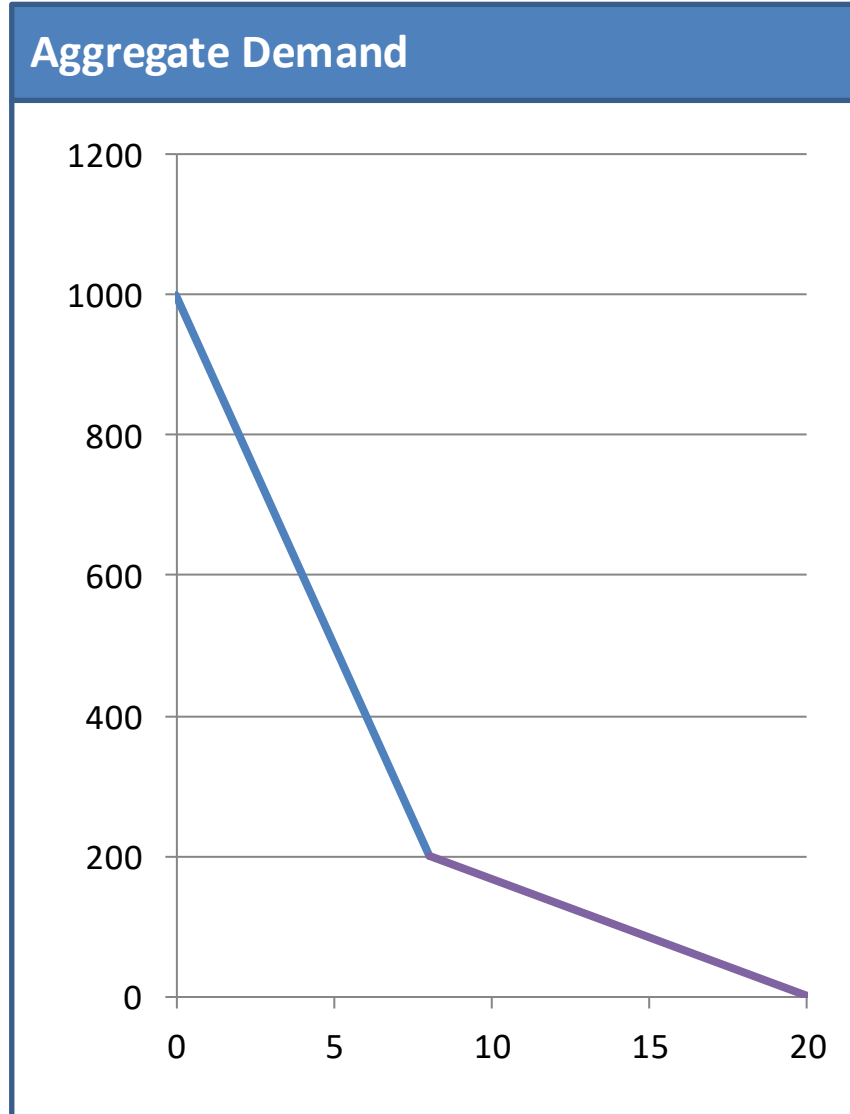
New Haven and Miami Demand



Aggregate Demand



Aggregating Demand: Customer Mix



Who are my customers?

- In this price range, I am only selling to New Haven-based MBAs.
- In this price range, I am selling both to Miami and New Haven-based MBAs.

Aggregating Demand

Important

When adding demand together, you always want to **add by quantity**, not price (i.e. be sure you have “demand,” not “inverse demand”).

Step 1

Write demand in terms of quantities:

- Boston Demand: $P = 1000 - 100Q_N \rightarrow Q_N = 10 - 0.01P$ (for $P \leq 1000$)
- Miami Demand: $P = 200 - 20Q_M \rightarrow Q_M = 10 - 0.05P$ (for $P \leq 200$)

Step 2

Add together over the appropriate range, and then (optionally) rewrite so price is on the left again:

- Total quantity as a function of price:

$$Q_{TOT} = Q_N + Q_M = 20 - 0.06P \text{ (for } P \leq 200\text{)}$$

$$Q_{TOT} = Q_N = 10 - 0.01P \text{ (for } 200 \leq P \leq 1000\text{)}$$

Where does this price range come from? It's the price where the Miami demand is zero.

- (Inverse) Demand is therefore

$$P = 333 - 16.7Q \text{ (for } 8 \leq Q \leq 10\text{)}$$

$$P = 1000 - 100Q \text{ (for } Q \leq 8\text{)}$$

Aggregating Demand: When Do We Do This?

Examples of when firms have to aggregate demand from different types of customers are all around us:

Situation	Types of Customer
New Haven Hotels	• Business travelers vs leisure travelers • They also vary in demand for weekday vs weekend
Costco	• Demand comes from “commercial” customers (bars and restaurants), as well as individual consumers.
Local Museums	• Demand from locals differs from demand among tourists.
Electric Utilities	• Demand patterns differ for industrial, commercial and residential customers.

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4. Questions and discussion

- **Application to TSA PreCheck**

Article Discussion: “Interested in TSA PreCheck?...”

Hypothetical Situation

Suppose it is the year 2014 (TSA PreCheck just introduced) and you are a regular traveler through a major airport.

Option 1

Wait in regular security line (LGA is #2 for longest TSA wait times)

Option 2

Pay fee for TSA PreCheck: cuts wait times to less than 10 min.

- What is your reservation price for TSA PreCheck?

How does your reservation price change if...

- in 2015, falling oil prices make air travel cheaper?
- in 2015, TSA begins randomly diverting some travelers to TSA PreCheck for free?
- in 2016, increased air travel and decreased TSA staff make regular lines longer?

Article Discussion: “Interested in TSA PreCheck?...”

Now consider...

Option 3

Amtrak offers high speed Acela service to Boston, one of your frequent routes.

How does this affect your reservation price for TSA PreCheck?

What if...

- Amtrak offered high-speed wifi (that worked well)?
- increased TSA PreCheck adoption made application wait lines longer?
- TSA PreCheck began requiring taking off shoes?



New, state-of-the-art, high-speed trainsets are coming to the United States. Building on Amtrak's expertise as the only high-speed rail operator in the U.S. and Alstom's record of delivering world class, proven, high-speed trainsets—Amtrak and Alstom have partnered to introduce the next generation *Acela*. These new trainsets will redefine the customer experience on Amtrak's Northeast Corridor.

Amtrak's new *Acela* fleet, scheduled to enter service on the Northeast Corridor (NEC) in 2023, will upgrade the travel experience for millions of passengers and set the stage for the next generation of train travel in America and on the Northeast Corridor via enhancements in comfort, technology, innovation, and safety on Amtrak's most environmentally sustainable fleet of trains to date. Additional features include:

- Comfortable seating with personal outlets, USB ports and winged headrests to serve as a barrier between the customer and his/her seatmate
- Spacious restrooms with contactless features and automatic door access
- Onboard information system providing real-time information such as train location, train speed, and conductor announcements
- Contactless storage option for luggage
- Café Car enhancements including convenient self-service options
- Accessibility features for people with disabilities



Project Summary

Timeline: The first trainset prototype began testing in 2019. In 2020, testing began along the Northeast Corridor. The new trainsets will enter revenue service in 2023.

Funding: RRIF Loan backed by NEC revenues

Partners: US DOT Federal Railroad Administration and Alstom

Factors Affecting your Reservation Price

A consumer's Reservation Price for a good or service depends on the characteristics of the good, as well as the prices and characteristics of substitutes and complements.

TSA will no longer require all passengers to take shoes off at airport security checkpoints



By Pete Muntean and Alexandra Skores, CNN

🕒 2 min read · Updated 5:38 PM EDT, Tue July 8, 2025



125 comments



Washington (CNN) — After nearly two decades, passengers going through airport security in the United States will no longer have to take their shoes off.

The Transportation Security Administration will be eliminating the security requirement “effective immediately,” Kristi Noem, the Secretary of Homeland Security, announced Tuesday.

Article Discussion: “Interested in TSA PreCheck?”

Challenge: fewer people than expected are signing up.

- Translation: reservation prices are less than the price.

Solution?

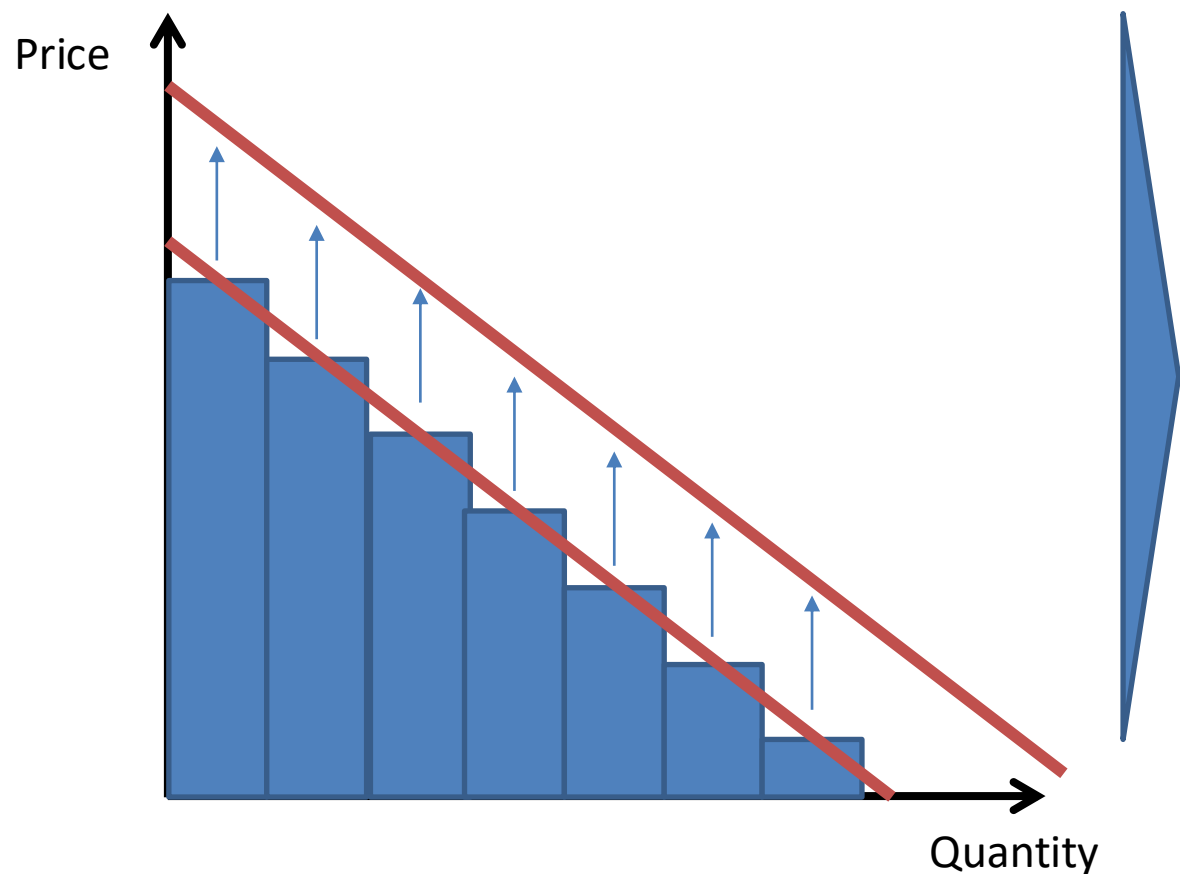
- Making it easier to sign up (at the airport, mobile enrollment)
- Shorter time horizon (from 5 years)
- Payment plan
- Other ideas?

Reservation prices and demand

Firms trying to sell products need reservation prices to be above purchase prices. Two ways to increase sales are lower prices (which firms don't like) or increase what consumers are willing to pay.

Article Discussion: “Interested in TSA PreCheck?”

How are these changes in reservation prices reflected in demand curves?



- Changes in reservation prices will move the demand curve around.
- We say that the changes *shift and/or rotate demand*.
- The new demand curve tells you the quantity that can be sold at a given price for the new reservation prices.

Firms Want to Change Your Reservation Price

Firms want you to be willing to pay more for their products. How do they raise your reservation price?

- The obvious answer? Improve the quality!
- The “easier” answer?
 - Marketing!

A great example: have you ever ordered “Patagonian toothfish” at a restaurant?

How about “Chilean Sea Bass”?



... they are the same thing. Reservation prices for the latter are higher!

Summary and Direction

1. Reservation Prices

- The *most* a consumer is willing to pay for a good or service

2. The Demand Curve and Consumer Surplus

- Reflects the *distribution* of reservation prices in a market: at a given price, how many are willing to buy?
- Consumer surplus increases as prices fall, since consumers are better off.

3. Aggregating Demand

- When adding demand from different types of consumers, add in *quantities*.

Next Lecture

- Supply curve: how much a firm is willing to produce at a given market price.
- Market equilibrium: putting demand and supply together.